frontiers in Ecology and Evolution

TYPE Brief Research Report
PUBLISHED 30 June 2026
DOI 10.3389/fevo.2026.1763394

Check for updates

OPEN ACCESS

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RECEIVED 08 December 2025
REVISED 20 May 2026
ACCEPTED 26 May 2026
PUBLISHED 30 June 2026

CITATION
Osterhaus DM, Salas AJ, Cutler PL, Weisenberger ME, Mower K, Lawson AJ, Wright TF, Fair JM and Desmond MJ (2026) Mass mortality of avian migrants in New Mexico, USA, that coincided with an extreme weather event.
Front. Ecol. Evol. 14:1763394.
doi: 10.3389/fevo.2026.1763394

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Mass mortality of avian migrants in New Mexico, USA, that coincided with an extreme weather event

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Many birds are migratory, and this life history strategy allows for maximized access to seasonally abundant resources and favorable climates. However, migration exposes birds to threats and stressors, resulting in high mortality during migration. Anthropogenic landscape alterations and climate change have intensified threats, and mass mortality events linked to extreme weather are more common in recent decades. Documenting mass mortality is critical for predicting future occurrences and implementing effective conservation. Here, we describe a mortality event that occurred throughout New Mexico, USA in fall 2020. Carcasses began appearing in the region in mid-August, during a period of extreme heat and drought. Following an extreme cold weather event on September 8–9th, the number of carcasses increased dramatically and expanded throughout the state. In total, we collected 628 carcasses comprising 58 species within Doña Ana and Otero Counties in New Mexico. Necropsy determined emaciation was the cause of death for 74.6% of carcasses. Live birds captured during the period of peak mortality (n = 223) were in similarly poor condition. This event provides a striking example of how multiple types of extreme stressors, in this case widespread drought and unseasonal cold, coincided with a mortality event, indicating a possible synergistic relationship between these factors and the mass mortality. Mortality events are likely to increase in frequency with intensifying climate change. Establishment of networks of biologists and researchers could improve our ability to identify and communicate developing mortality events, organize data collection, and improve understanding of the causes and consequences of mortality.

KEYWORDS
avian migration, bird mortality, climate change, extreme weather, mass mortality, migration mortality

1 Introduction

Billions of birds migrate seasonally, traveling hundreds to thousands of kilometers, often crossing continents or hemispheres to reach wintering and breeding locations. Worldwide, 19% of avian species are migratory and in North America, 70% of terrestrial species migrate (Rappole, 1997; Kirby et al., 2008; Horton et al., 2019). Migration allows birds to exploit seasonally abundant resources and avoid unfavorable climates, but it comes with substantial risks. Migratory birds encounter numerous hazards as they move across the landscape. Predation risk is high as birds navigate unfamiliar stopover sites and weather can disrupt migration progression. Birds must cross large expanses of inhospitable habitat including oceans, mountain ranges and deserts. Exposure to disease is increased, and birds may encounter reduced foraging opportunities (Altizer et al., 2011; Chapman et al., 2014; Fudickar et al., 2021). The added novel stressors of anthropogenic activities and climate change exacerbate these hazards, and mortality is higher for many species during migration compared to other periods of the annual cycle as a result (Sillett and Holmes, 2002; Klaassen et al., 2014; Newton, 2025).

During migration, birds are constantly balancing mortality risk against fitness benefits and are pushing physiological limits (Alerstam, 2011; Newton, 2025). Climate change and anthropogenic alteration of the landscape have increased the variability of conditions that migrants experience, further intensifying the difficulty of migration. Land conversion, wildfires, and droughts reduce the availability and quality of stopover sites, limiting refueling opportunities (Kittelberger et al., 2022). Artificial light disrupts migration, causing birds to deviate from preferred routes, increasing the risk of collision with buildings, forcing stopovers in suboptimal habitats, and leading to excess energy expenditure (Loss et al., 2015; Van Doren et al., 2017; Osterhaus et al., 2025a). Ongoing shifts in primary productivity phenology creates spatial and temporal mismatches between migrants and the resources they depend on to survive migration (Jones and Cresswell, 2010; Saino et al., 2011; Senner et al., 2017). Extreme weather events are becoming more common in recent decades, exposing migrants to adverse, and often unsurvivable weather conditions (Stott, 2016; Cohen et al., 2021). In recent years, zoonotic disease outbreaks have caused mass mortality events in a variety of species (Avery-Gomm et al., 2024; Knief et al., 2024). Occasionally, multiple stressors coincide in such a way to trigger mass mortality events—sudden, large-scale die-offs of birds in which fatalities substantially exceed baseline mortality rates (Newton, 2007). In birds, mass mortality events occur within both resident and migratory populations, can impact all life stages and influence the long-term evolutionary trajectories of populations (Quintana et al., 2022; Stager et al., 2026), and have increased in frequency and magnitude in recent decades (Fey et al., 2015). Newton (2007) summarized 56 mass mortality events for migratory birds associated with weather since the late 1800s; most occurred during spring migration or just after arrival on the breeding grounds and were associated with extreme or unseasonal weather. Periods of cold after arrival on breeding grounds have resulted in mortality events associated with starvation of invertivorous birds and reports of carcasses along roadsides and around human habitations (Manville, 1957; Marcstrom and Mascher, 1979). For example, in March of 1904, 750,000 Lapland longspurs (*Calcarius lapponicus*) died within a 2.6 km^2 area because of a late season snowstorm in Minnesota, USA; with estimated mortality up to 1.5 million over a larger area; some birds were attracted to lights in town resulting in collisions with anthropogenic structures (Roberts, 1907a, Roberts, 1907b). In a more recent example, thousands of avian migrants died in and around Chicago, Illinois, USA when extreme weather and anthropogenic structures created hazardous conditions for migrants moving through the region (Farnsworth et al., 2024). Mass mortality events do not always impact juveniles and adults evenly (Piatt et al., 2020; Quintana et al., 2022; Solís et al., 2024). For example, in 2016, 86.5% of Imperial Cormorant (*Leucocarbo atriceps*) chicks died as the result of a heatwave, while no adult mortalities were discovered (Quintana et al., 2022). Finally, while many mortality events are associated with cold weather, in recent years multiple mass mortality events have been caused by extreme heat (Piatt et al., 2020; Quintana et al., 2022).

The precipitous decline of avian populations has placed many species at heightened risk of extinction (Rosenberg et al., 2019). For some species, mass mortality events may represent critical tipping points beyond which small populations are unable to recover. From 2014-2016, a heatwave led to the loss of one quarter of the Common Murre (*Uria aalge*) population breeding in the Gulf of Alaska and southeast Bering Sea, and 22 complete reproductive failures across multiple colonies (Piatt et al., 2020). In 2022, a catastrophic breeding failure occurred in Emperor Penguins (*Aptenodytes forsteri*) as the result of sea ice contraction (Fretwell et al., 2023). Beyond the initial mortality, these events can have long-lasting impacts on the ecology and evolution of impacted populations (Stager et al., 2026). The frequency and intensity of events such as extreme weather (Stott, 2016; Jain et al., 2022), large-scale wildfires (Mansoor et al., 2022), and heat-waves (Meehl and Tebaldi, 2004) are increasing globally, amplifying the threat to vulnerable bird populations. However, mass mortality events can be difficult to predict, detect and accurately quantify, as migration routes cover vast geographic areas making it impossible to search completely, and baseline mortality rates are rarely known. Systematic documentation of mass mortality events and associated environmental and biological factors related to them will enhance our ability to predict the occurrence of these events and improve our understanding of how these events affect populations.

Here, we provide a descriptive report of a migratory bird mass mortality event that occurred in southern New Mexico, USA throughout the fall migration season of 2020. The event was preceded by a summer dominated by extreme drought and widespread wildfires across western North America that contributed to abnormally poor air quality. Initial documented mortalities were concentrated on White Sands Missile Range (WSMR), a known stopover site in southern New Mexico, but mortality became widespread throughout New Mexico and in neighboring states following an extreme weather event in which temperatures dropped far below seasonal norms, accompanied by strong winds and precipitation. We describe the environmental conditions that coincided with and may have contributed to this mortality event, report on the species and demographic composition of carcasses, detail

necropsy findings including tests for potential toxicants and diseases and describe body condition of live birds captured during the event. Additionally, as our data collection efforts were reactive and unstandardized, which precluded robust statistical analyses, we discuss how increased preparedness for coordinated data collection efforts for future events would facilitate more robust analyses of the causes of these events and allow for improved enumeration of total mortality and impacts to populations.

2 Materials and methods

2.1 Weather and smoke data

We obtained daily temperature (minimum, maximum, and mean) and precipitation data for Doña Ana County (32.2816, −106.7522) and Otero County (32.3731, −106.3785) from the PRISM climate data platform (PRISM Group, 2025). To contextualize conditions during the mortality event, we compared 2020 values to historical climate normals (30-year norms from 1991–2020) for the same locations and time of year. Ambient wildfire smoke data were obtained from Childs et al. (2022). Weather (temperature and precipitation) and smoke data were averaged between the two counties for analyses. All data preparation and analyses were conducted in R version 4.5.0 (R Core Team, 2025).

2.2 Carcass collections

Carcass collections occurred in Doña Ana and Otero Counties in southern New Mexico (Figure 1). Doña Ana and Otero counties are within the Chihuahuan Desert ecoregion with a 20-year normal annual mean temperature ranging from 17.1–19.0°C (PRISM Group, 2025). Annual precipitation ranges from 146–410 mm, and most rainfall occurs during the late summer monsoon (PRISM Group, 2025). Much of our data collection occurred in the southern portion of WSMR within Doña Ana County. WSMR is situated with Las Cruces to the west (population ~105,000), El Paso to the south (population ~680,000), and Alamogordo to the east (population ~33,000). Land cover types outside the urban and agricultural landscapes include desert scrub or disturbed desert scrub dominated by honey mesquite (*Prosopis glandulosa*), soaptree yucca (*Yucca elata*), and creosote bush (*Larrea tridentata*). To the west of WSMR, the Rio Grande corridor is largely comprised of riparian species, including willows (*Salix* spp.) and non-native tamarisk (*Tamarix* spp.) that are known to provide crucial stopover habitat for migrants (Kelly et al., 2002; Walker, 2008).

Carcasses were opportunistically collected on WSMR from September 2nd through October 28th, 2020. The opportunistic nature of our carcass collections precludes robust statistical analyses, and consequently our results must be considered descriptive. All carcasses and partial remains were placed in bags and labeled with date and location and stored in a −20°C freezer. Carcasses were identified to species, aged (hatch-year or after-hatch-year) and sexed (Pyle, 1997), and assigned to a foraging guild following Dubovyk (2024). A subset of carcasses collected on WSMR and others collected opportunistically throughout New Mexico on overlapping dates that were in good condition (little to no evidence of decomposition) were sent to the U.S. Fish and Wildlife Service (USFWS) Forensic Laboratory (Ashland, OR) and the U.S. Geological Survey (USGS) National Wildlife Health Center (Madison, WI) for necropsy in which radiographic, external, internal, and microscopic examinations were completed to determine cause of death. Carcasses were tested for organic toxicants, minerals and heavy metals, and various diseases that have been linked to previous mass avian mortality events (Fey et al., 2015). Organic toxicants, including drugs, pesticides, and industrial compounds were tested for via gas chromatography and mass spectrometry of the liver (Buchweitz et al., 2018). Minerals and heavy metals (including arsenic, cadmium, cobalt, copper, iron, lead, manganese, mercury, molybdenum, selenium, thallium, and zinc) were tested for via inductively coupled plasma mass spectrometry.

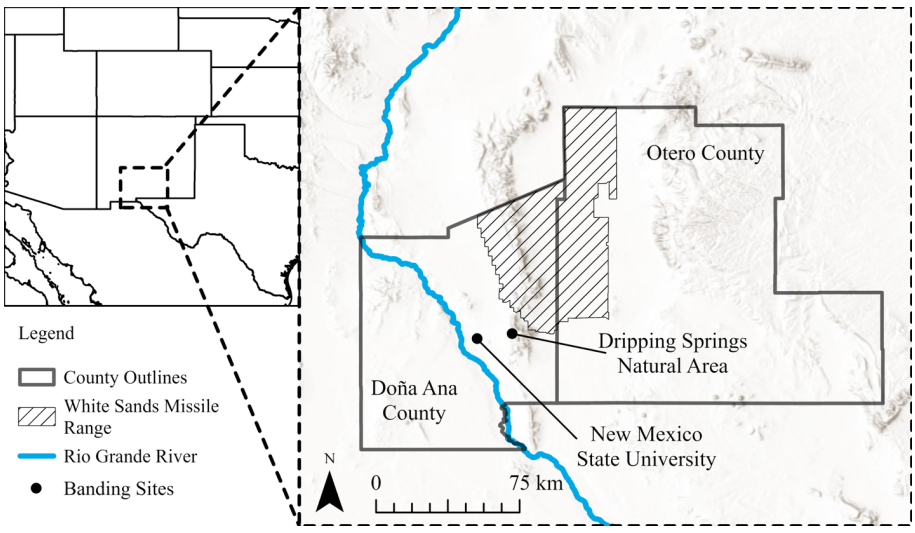


FIGURE 1
Map of Doña Ana and Otero Counties in southern New Mexico, USA, highlighting the banding sites and White Sands Missile Range.

Avian influenza, avian paramyxovirus type 1 (Newcastle disease virus), and herpesvirus were tested for by reverse transcription polymerase chain reaction using tracheal and cloacal swabs.

2.3 Live bird captures

To assess species and demographic composition, and body condition of migrants during the mortality event, we captured birds at two locations in Doña Ana County between September 11th–17th, 2020. Five mist nets were placed at New Mexico State University (NMSU) main campus along an agricultural field border (32.279970, −106.758679) (Figure 1) on September 11th–13th and 15th–17th between the hours of 7:00–10:00 and 17:00–19:30. Additionally, five nets were placed at Dripping Springs Natural Area within Organ Mountains-Desert Peaks National Monument (32.329844, −106.589531) (Figure 1) on September 12th and 15th from 7:00–20:00. Each captured individual was fitted with a USGS leg band, weighed (g), aged (hatch-year or after-hatch-year), sexed (Pyle, 1997), and visually scored for body fat (from 0 to 5, 0 being no fat and 5 being excessive fat) prior to release (Fair et al., 2023). All birds were handled following the methods approved by New Mexico State University Institutional Animal Care and Use Committee (permit #2020-029).

3 Results

3.1 Regional climate and extreme weather coinciding with the mass mortality event

During summer and early fall 2020, much of the western United States was experiencing long-term drought with parts of nine states including New Mexico experiencing extreme to exceptional drought (0 to 5th percentile of historical moisture; NIDIS and NOAA, 2020). These conditions resulted from low snowpack during winter 2019–2020, below-normal monsoonal rainfall, and record-high August heat (NIDIS and NOAA, 2020). Hot, dry conditions contributed to an unusually strong fire season, with more than 3,000 wildfires (20% above the 10-year average) that burned over 420,873 ha in the western US (72% above the 10-year average) (NICC, 2020). Fires across the western United States contributed to abnormally poor air quality during the mortality event and throughout fall migration (Yang et al., 2021). From 2006 to 2019, in the month of September, mean daily wildfire smoke PM2.5 averaged between Doña Ana and Otero Counties was 0.13 (μg/m³) and ranged from 0 to 0.96 (μg/m³). In 2020, mean daily wildfire smoke PM2.5 averaged between Doña Ana and Otero Counties was 3.37 (μg/m³) (Figure 2C).

Beginning in mid-August 2020, small numbers of carcasses were reported from WSMR. Such reports continued into early September, which prompted coordinated, though unstructured, carcass collection efforts on WSMR. In early September, an intense storm system moved through Colorado and New Mexico. On September 8–9th the Denver metropolitan area underwent a rapid shift from record high temperatures to unusually low temperatures due to a cold front from the north that brought one of the earliest snowfalls on record (2.5–25 cm) to the region. The storm front continued south into northern New Mexico, with snow and winds exceeding 112 km hr^{-1} recorded in Albuquerque. The southern New Mexico region, including Doña Ana and Otero Counties, experienced rainfall and sudden cold temperatures during this time, with an average temperature of 27.05°C recorded the day before the front moved through (September 8th), dropping to 13.20°C by September 10th. Precipitation totaling 13.49 mm fell on September 9th and 10th (Figures 2A, B; PRISM Group, 2025). Mortalities prior to the extreme weather event were documented only on WSMR, and not reported in other areas of the state, although biologists across the state were aware of the mortalities at WSMR and were encouraged to be vigilant. During and immediately following the extreme weather event, mortalities were documented throughout the state, and carcasses were reported and collected through October 28th.

3.2 Collected avian carcasses and necropsy findings

We collected 628 carcasses comprising 58 species in Doña Ana and Otero Counties between September 2nd and October 28th, 2020 (Supplementary Table 1). The number of carcasses collected each day peaked during and immediately following the extreme weather event between September 8th–12th, with a maximum of 162 carcasses collected on September 10th (Figure 2D). The timing of mortalities for hatch-year and after-hatch-year birds was similar (Figure 2E), and hatch-year birds accounted for 69% of the carcasses that could be assigned to an age class mirroring the expected age structure of the migrant community in this region at the time of the event (Osterhaus et al., 2026). The sex ratio of carcasses was close to one-to-one, with 46% of known sex birds being female and 54% male. Most carcasses collected were invertivores (64%), with collection of invertivore carcasses peaking during and immediately after the extreme weather event (Figure 2F). Nine species accounted for 75% of carcasses collected, with western wood-pewee (*Contopus sordidulus*) being the most common (Figure 3; Supplementary Table 1). Other common species included Wilson's warbler (*Cardellina pusilla*), chipping sparrow (*Spizella passerina*), yellow-headed blackbird (*Xanthocephalus xanthocephalus*), violet-green swallow (*Tachycineta thalassina*), lark sparrow (*Chondestes grammacus*), and brown-headed cowbird (*Molothrus ater*) (Figure 3; Supplementary Table 1). Mortalities reported from WSMR prior to the extreme weather event were comprised primarily of granivores, dominated by yellow-headed blackbirds, brown-headed cowbirds, and lark sparrow; however, this was not captured well in our dataset because coordinated carcass collection was not initiated until September 2nd (Figure 2F; Cutler pers. comm.).

Necropsies were completed for 59 carcasses comprising 27 species, with 16 carcasses originating from WSMR and the remaining 43 from throughout New Mexico (Supplementary Table 3). Based on the laboratory examiner's notes, cause of death was emaciation-related for 74.6% of the necropsied carcasses, with 15.3% of deaths attributed to other causes (e.g., trauma), and 10.2% undetermined (Figure 4A; Supplementary Table 3). Within guilds, cause of death was emaciation-related for 75% of granivores

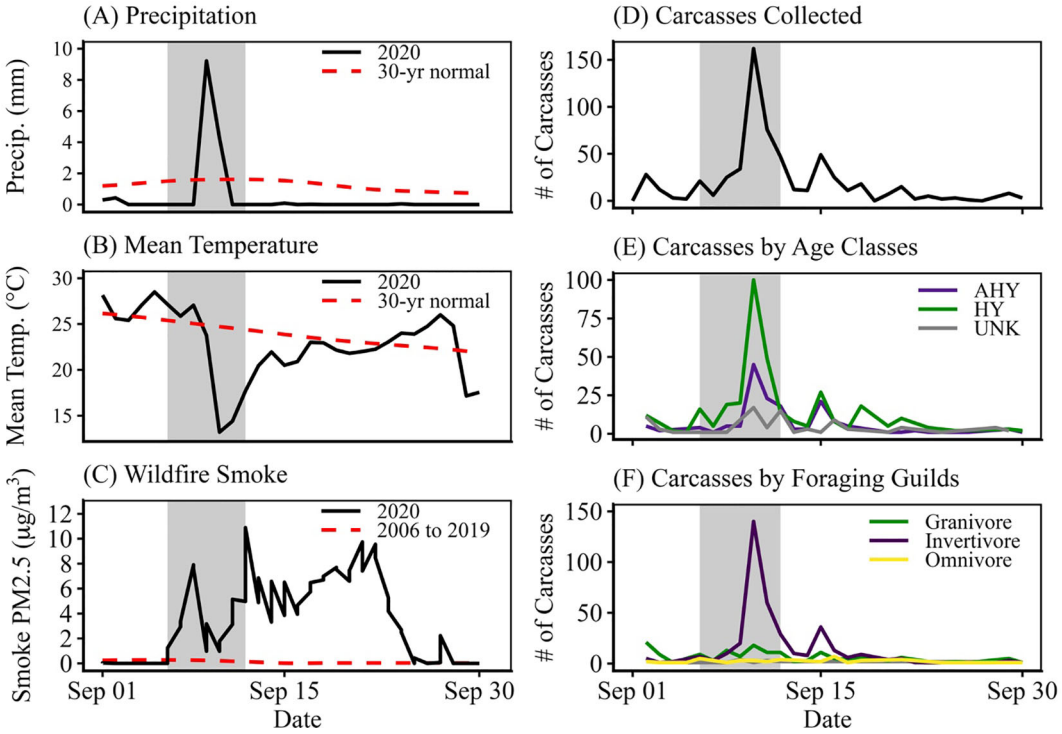


FIGURE 2
Summary of the avian mortality event and associated environmental conditions from September 1st – 30th, 2020, showing **(A)** precipitation and **(B)** mean temperature along with 30-yr normals, and **(C)** wildfire smoke (the dashed red line represents average daily smoke PM2.5 from September 1st–30th for 2006 to 2019, and the solid black line is the daily smoke PM2.5 for 2020), **(D)** carcasses collected, **(E)** carcasses collected by age class (AHY is after-hatch-year, HY is hatch-year, UNK is unknown), and **(F)** carcasses collected by foraging guild. Weather and smoke data were averaged between Doña Ana and Otero Counties, New Mexico, USA. Carcass data is for all carcasses collected in Doña Ana and Otero Counties only. The gray shaded region highlights the severe weather event occurring from September 6th–12th, 2020.

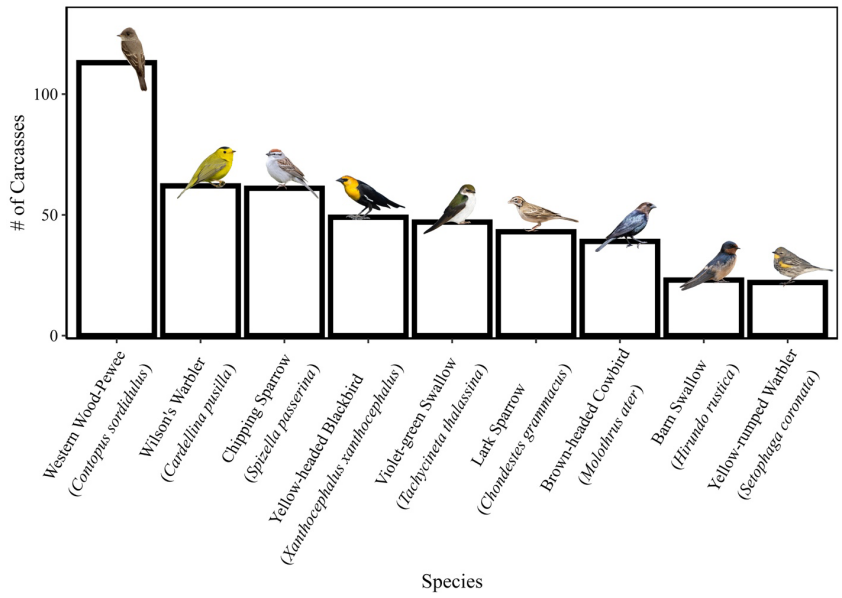


FIGURE 3
Frequency of carcasses by species for each species with > 20 carcasses collected during the fall 2020 avian mortality event in Doña Ana and Otero Counties, New Mexico, USA. Species images were obtained with permission from the Cornell Lab of Ornithology and Macaulay Library.

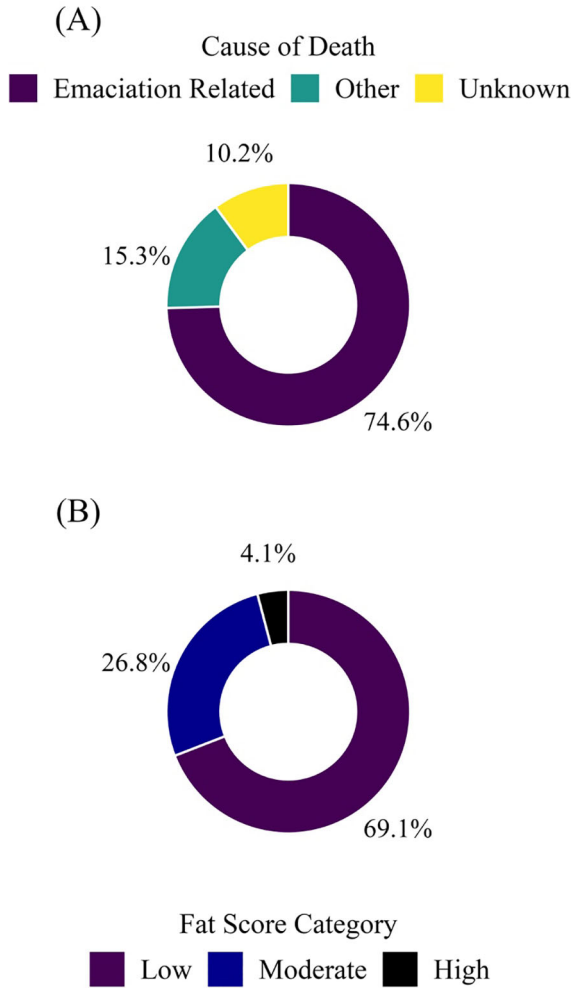


FIGURE 4
Comparison between **(A)** suspected cause of death based on necropsy results for 59 carcasses collected during the mortality event in New Mexico and necropsied by the USFWS National Fish and Wildlife Forensic Laboratory, Ashland, Oregon, USA and USGS National Wildlife Health Center, Madison, Wisconsin, USA, and **(B)** fat scores for 220 live birds captured in Doña Ana County, New Mexico, USA during the fall 2020 avian mortality event with scores categorized by low (scores of 0 or 1), moderate (2 or 3), and high (4 or 5).

(n = 8 total individuals necropsied), 74% of invertivores (n = 46), 100% of nectivores (n = 2), and 66% of omnivores (n = 3). Birds with deaths attributed to emaciation had markedly depleted fat stores, significant loss of muscle mass, pronounced sternal keels, and lack of ingesta in the upper gastrointestinal tract, indicating prolonged starvation. No birds tested positive for organic toxicants, minerals/heavy metals, avian influenza, avian paramyxovirus, or herpesvirus. While not implicated as a cause of death, irritation of the respiratory system was noted by the examiner for multiple individuals and was thought to be caused by inhalation of wildfire smoke.

3.3 Live bird captures and atypical behavioral observations

A total of 223 live birds were captured at NMSU and Dripping Springs Natural Area. The age ratio of captured birds (63% hatch-year, 37% after-hatch-year) was similar to that of collected carcasses. Sex ratios were also similar between captured birds and collected carcasses, with captured birds being 48% female and 52% male. Fat scores were low for captured birds with 69.1% having fat scores of 0 or 1, closely mirroring the proportion of mortalities attributed to emaciation in the necropsies (Figures 4A, B). Species assemblage of the live birds mirrored that of the collected carcasses, with the exception that western wood-pewee were infrequently captured, but were commonly observed in the habitats near nets (Supplementary Table 2).

During and immediately following the extreme weather event, we observed and received reports of several atypical behaviors by migratory birds including swallows (family Hirundinidae) grounded, lethargic, and dying on golf courses, western wood-pewees roosting on human structures (porches and patios), birds foraging in roadways, and cowbirds and blackbirds grounded, lethargic and dying on manicured lawns. Other behaviors reported by members of the public during the extreme weather included barn swallows (*Hirundo rustica*) and violet-green swallows (*Tachycineta thalassina*) jointly crowding into a single old barn swallow nest before dying, individuals of several invertivorous species entering buildings where they were later found dead, birds hitting windshields at night, and western bluebirds (*Sialia mexicana*) crawling into burrows in the ground where they later died. Numerous reports were received concerning birds being hit by cars because they were too lethargic to fly off the ground.

4 Discussion

In the fall of 2020, we documented a mass avian mortality event in New Mexico, that resulted in hundreds of documented deaths across 58 species. Large numbers of mortalities were reported throughout the state following an extreme weather event in early September that coincided with peak passerine migration through New Mexico (Osterhaus et al., 2025b). In southern New Mexico, observed mortalities occurred over a longer time span than other parts of the state and region, with carcasses appearing on WSMR in mid-August and continuing to appear through the end of October. Early observed mortalities on WSMR suggest migrants were already in poor condition prior to the extreme weather event, possibly related to long-term widespread drought that had deteriorated the quality of stopover habitat and food resources available throughout western North America. The coincidence of the extreme weather event on September 8–9th with the increase in carcasses collected shortly thereafter, is consistent with the hypothesis that extreme weather synergistically interacted with birds in poor body condition and may have resulted in this widespread mortality event. The possible explanation for this mass mortality event is corroborated by necropsy results that revealed many birds exhibited strong indications of emaciation and long-term starvation including depleted fat reserves, severely shrunken flight muscles, and empty stomachs (New Mexico Department of Game and Fish, 2020; Kittelberger et al., 2022). Likewise, live birds captured in the area during the time when mortalities peaked also showed generally

poor body condition and low to no fat reserves. Notably, there was no evidence for the involvement of acute effects of wildfire, nor toxicants or disease, with all necropsied birds negative for those agents tested. However, West-Nile Virus was not tested for, and could be tested for in future mortality events as it has been implicated in previous large-scale avian mortality events.

Unseasonable cold weather during migration can be catastrophic for birds, particularly invertivorous species. During migration, invertivores may be more sensitive to reduced food resources than species of other foraging guilds (Carlisle et al., 2009). Several studies have documented starvation during unseasonable cold weather with migrants, primarily invertivores, in emaciated conditions and lacking fat reserves (Ligon, 1968; Whitmore et al., 1977; Marcstrom and Mascher, 1979). Unseasonable cold weather during spring migration has resulted in mass mortalities of western birds similar to that which we report here for fall 2020 (Ligon, 1968; Whitmore et al., 1977). In Utah, Whitmore et al. (1977) documented the mortality of 32 species of birds, including many of the same species we reported in New Mexico. Similar to New Mexico, Whitmore et al. (1977) found birds with empty stomachs, atrophied flight muscles, and no subcutaneous fat and attributed mortality to starvation or freezing conditions.

While unseasonably cold weather may be particularly detrimental for invertivores, extreme heat and long-term drought can be detrimental to birds in general (Piatt et al., 2020; Quintana et al., 2022). In southern New Mexico, mortality of granivores began in mid-August during extreme heat and widespread and intense drought. Similar to invertivores, granivorous migrants are constrained by food availability and vegetation green-up, and have earlier migration and poorer body condition during years when temperatures are above normal (Dunning, and Brown, 1982; Renfrew et al., 2013; Miller et al., 2015). Extreme drought was widespread in New Mexico and across the interior regions of the western U.S. in 2020 (Zhang et al., 2021). These conditions appeared to contribute to the early mortalities of migrants documented on WSMR up to three weeks prior to the widespread mortality across the state. Similarly, in 2023, a mortality event involving primarily granivorous migrants occurred on WSMR, also associated with extreme heat and drought (Boland, 2024). Additionally, extreme wildfire activity widespread throughout the western U.S. in 2020 may have exposed birds to sublethal effects of toxins via smoke during migration, and interfered with migration timing and routes, and selection of stopover sites (Overton et al., 2022). Multiple necropsied carcasses were found to have respiratory system irritation consistent with exposure to wildfire smoke, but this was not implicated as a cause of death. Future work continuing to assess how the interaction between avian migrants and wildfire smoke affects migratory routes, decision making, and survival would be helpful.

The mass mortality event in New Mexico and its probable link to coinciding extreme weather events highlights the negative consequences of multiple stressors, each of which are expected to occur more frequently with intensifying climate change. During the summer breeding season of 2020, birds throughout the western U.S. experienced degraded habitat conditions intensified by long-term drought. It is possible that migrant species in the west were already in poor condition as they began their southward migration, as evidenced by long-term starvation being the cause of death in 76% of necropsied carcasses. As these birds migrated south, they would experience extreme heat, large wildfire smoke plumes, and degraded habitats in stopover sites, where replenishing depleted energetic stores would likely be difficult. As such, early mortalities documented by this study, like the granivores found dead on WSMR in August 2020, could be a reflection of cumulative, chronic stressors that left birds in poor condition more vulnerable to mortality. The massive extreme weather event that moved through the southwest region during peak migration, may have acted as an acute stressor that pushed migrants already in poor body condition, past their physiological limit, resulting in widespread mortality.

Mass mortality events have the potential to exacerbate on-going population declines across a range of species. Even in common species, many of which have experienced widespread declines in North America (Rosenberg et al., 2019), mortality events can have significant population-level impacts. Populations of western North America forest birds, which represented a large portion of the carcasses we collected, are experiencing ongoing declines, having declined by 29.5% overall since 1970 (Rosenberg et al., 2019). A mortality event in Europe in 1974 was estimated to cause the deaths of millions of swallows and martins and was associated with an estimated 25–30% decline in house martin (*Delichon urbicum*) populations in Switzerland (Newton, 2007). While the mortality event described here occurred after the breeding season, events that occur during the nesting season may result in reproductive failure for entire populations which for short-lived species may be particularly detrimental (Piatt et al., 2020; Quintana et al., 2022; Solís et al., 2024). Future work that assesses the capacity of populations to recover following a mass mortality event may help identify priorities for conservation efforts to mitigate the effects of such events in the future.

Though drought patterns and weather forecasts may be helpful indicators, predicting the timing of mass avian mortality events is challenging. As such, data collection efforts during a mortality event are often hampered by a lack of organization and prior planning. Improved early detection of events and coordinated preparedness to respond to events are needed. During the mass mortality event of fall 2020 in New Mexico, biologists from multiple agencies and organizations were able to communicate from a pre-established network, the New Mexico Avian Conservation Partners, to organize and coordinate data collection efforts (New Mexico Avian Conservation Partners, 2025). Despite this advantage, it was challenging to mobilize a well-planned study with standardized sampling to document mortality across the state. Our collection of carcasses was opportunistic, early mortalities were not well documented, data were not collected via standardized or replicated surveys to estimate mortality, and there were no existing data to establish baseline levels of mortality. To enhance preparedness and response to future events, there is a need to establish an engaged network of conservation practitioners and researchers equipped to identify when a mortality event is occurring, and serve as a centralized point of contact for reporting mortalities and coordinating data collection. Such networks have aided in data collection during past mortality events (Stager et al., 2026), and could further

build scientific capacity by establishing best practices for carcass survey design to maximize mortality estimate precision and accuracy (e.g., Boland et al., 2026), and offer training opportunities for carcass surveys.

Our analysis of the mortality event described herein was hindered by the opportunistic way in which data were collected. During future mortality events, rapidly deployed coordinated surveys covering affected areas systematically and including a range of habitats could maximize the quality and utility of data collected. The creation of a broad scale pre-established network of trained survey leaders and volunteers with assigned survey routes would allow for rapid deployment of surveys in response to an active mortality event. Creating survey routes in advance would ensure that all available habitat types are sampled proportionally, including areas with artificial light at night and those without, limiting biases in species composition of collected carcasses stemming from habitat preferences or attraction to light. In addition to surveys during an event, periodically surveying the pre-established survey routes using the same methodology outside of mortality events could also be useful for understanding baseline rates of mortality associated with migration and provide an important point of comparison for data from future mortality events. To improve the quality of data collected through carcass surveys, habitat characteristic data, relevant for carcass detectability, could be collected at the time of survey route establishment (and updated periodically) which would allow for more rapid analysis and more accurate estimates. Importantly, quantifying effort during surveys, including area searched, duration, number of observers, and environmental conditions, is also necessary to correct total mortality estimates for imperfect detection (Kéry and Schmidt, 2008; Barrientos et al., 2018; Boland et al., 2026; Huycke et al., 2026). Platforms such as ArcGIS Survey123 (Hennig et al., 2023) or Open Data Kit (Brunette and Hartung, 2023) could be used to expedite the collection of and transfer of data from surveyors to coordinators and analysts, and would ensure that data are collected in a standardized format across all surveys, a key consideration when many surveyors may be collecting data. Furthermore, training and testing surveyors prior to actual surveys can provide critical early feedback to surveyors and improve their performance (Linz et al., 1991; Ponce et al., 2010; Boland et al., 2026). For example, searcher efficiency or detection trials involve placing carcasses on survey routes without the surveyor's knowledge and comparing the number of carcasses detected with the number available to estimate surveyor performance, an important factor needed to correct for surveyor bias in carcass collection data (Boland et al., 2026). In addition to variation arising from the detectability of carcasses due to factors like survey conditions, substrate, and surveyor performance, accounting for removal of carcasses by scavengers can also improve mortality estimate precision and accuracy. Determining how long carcasses persist on survey routes before they are removed by scavengers allows for mortality estimates to be corrected for scavenger loss based on the time between the mortality event occurring, and when surveys were conducted (Huycke et al., 2026). If scavenger removal cannot be examined on the survey routes, using removal rates from studies in similar environments may allow for improved quantification of absolute mortality. Overall, proactively designing and coordinating surveys and collecting the data needed to correct for sources of bias ahead of time will allow for greatly improved estimates of total mortality that may be reasonably extrapolated beyond the survey routes that were sampled, allowing for broadscale quantification of a mass avian mortality event.

4.1 Conclusion

Migration involves an intricate balance of survival and future fitness. The mass avian mortality event documented here is a striking case of the threat that climate change poses for migrants, and an example of how multiple coinciding extreme weather events might tip the balance from survival to death for migrants, and result in large-scale mortality. Emaciation and poor body condition of both live migrants and collected carcasses indicate widespread issues in resource quality and availability, likely driven by long-term drought (Yang et al., 2021). The first mortalities documented in mid-August were migratory granivores. Then, as migration reached its peak, an extreme weather system brought unseasonably cold temperatures to the region, probably tipping the balance from survival to death for birds that were already in poor condition prior to the extreme weather. These weather events, both drought and unseasonable cold, may have acted synergistically to generate a mass mortality event. Extreme heat and long-term drought, as well as other forms of extreme weather like the cold front that created the peak in mortality in this event, are expected to occur more frequently and with greater intensity with continuing climate change (AghaKouchak et al., 2020; Irannezhad et al., 2022; Nemes et al., 2023); therefore, the risk of future mass mortality events is likely to increase.

To enhance preparedness and response to future mass mortality events, there is a need for a coordinated network of conservation practitioners, land managers, researchers, and volunteers using standardized protocols for detection, reporting, and data collection before, during, and after mortality events. Such a network could greatly increase our understanding of mass avian mortality events, the conditions that contribute to them, how they impact populations, and how they may be avoided or mitigated in the future.

Data availability statement

The original contributions presented in the study are publicly available. This data can be found here: Osterhaus, D. M., Salas, A. J., Cutler, P. L., Weisenberger, M. E., Mower, K., Lawson, A. J., Wright, T. F., Fair, J. M., and Desmond, M. J. (2026b). Data from: Mass Mortality of Avian Migrants in New Mexico, USA, that Coincided with an Extreme Weather Event. Fish, Wildlife, & Conservation Ecology: Datasets. 2. https://digitalcommons.nmsu.edu/fish-wildlife-conservation-data/2.

Ethics statement

The animal study was approved by New Mexico State University Institutional Animal Care and Use Committee. The study was conducted in accordance with the local legislation and institutional requirements.

Author contributions

DO: Data curation, Formal analysis, Investigation, Methodology, Project administration, Visualization, Writing – original draft, Writing – review & editing. AS: Conceptualization, Data curation, Investigation, Methodology, Writing – original draft, Writing – review & editing. PC: Conceptualization, Investigation, Resources, Writing – review & editing. MW: Data curation, Investigation, Methodology, Writing – review & editing. KM: Data curation, Investigation, Methodology, Writing – review & editing. AL: Data curation, Investigation, Methodology, Writing – original draft, Writing – review & editing. TW: Funding acquisition, Investigation, Methodology, Resources, Supervision, Writing – review & editing. JF: Conceptualization, Investigation, Methodology, Writing – review & editing. MD: Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing.

Funding

The author(s) declared that financial support was received for this work and/or its publication. Funding for this research was provided by the NMSU Natural Resources Career Track Program funded by USDA Hispanic Serving Institutions Program grant number 2015 38422-30947 (M.J. Desmond, P.I.), and the NMSU Avian Migration Program funded by the USDA Hispanic Serving Institutions Program grant 2022-77040-37638 (T.F. Wright, P.I.). This is a New Mexico Agricultural Experiment Station publication supported by state funds and the U.S. Hatch Act.

Acknowledgments

The authors are grateful to Erin Duvuvuei, Corinne Arthur and the New Mexico State University students that assisted on the carcass searches, live bird mist netting, and data collection. From the beginning of this event, it was collaborative effort between Federal and State agencies, universities, and other organizations across New Mexico and the general public. We thank Dr. Hon Ip and his diagnostic team at the U.S. Geological Survey (USGS) National Wildlife Health Center for necropsies and avian influenza testing and Delivan Roper, USFWS for coordinating necropsies at the USFWS Forensics Laboratory. We thank the Cornell Lab of Ornithology and the Macaulay Library for providing access to the bird images used in Figure 3, and the following contributors who submitted the images to the Macaulay Library, Mason Maron (barn swallow), Brad Imhoff (brown-headed cowbird), Ben Lucking (chipping sparrow), a contributor (lark sparrow), Bridget Spencer (violet-green swallow), Evan Lipton (western wood-pewee), Gerrit Vyn (wilson's warbler), Zach Westfall (Yellow-headed Blackbird), and Kellen Apuna (yellow-rumped warbler). Thank you to the reviewers and Erin K. Buchholtz who provided constructive reviews for our manuscript.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary Material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/fevo.2026.1763394/full#supplementary-material

SUPPLEMENTARY TABLE 1
Summary of carcasses collected by species showing counts by age classes (HY, hatch-year; AHY, after hatch-year; UNK, unknown) and sex (F, female; M, male; UNK, unknown) of carcasses collected from the fall 2020 mortality event in Doña Ana and Otero Counties, New Mexico, USA.

SUPPLEMENTARY TABLE 2
Summary of live birds captured by species showing counts by age classes (HY, hatch-year; AHY, after hatch-year; UNK, unknown) and sex (F, female; M, male; UNK, unknown) of birds captured at New Mexico State University and Dripping Springs Natural Area in New Mexico, USA from September 11th–17th, 2020.

SUPPLEMENTARY TABLE 3
Summary of necropsy results and collection dates and locations for the 59 bird carcasses necropsied from the 2020 avian mortality event in New Mexico, USA. Necropsies were completed by either the U.S. Fish and Wildlife Service Forensic Laboratory (Ashland, OR) or the U.S. Geological Survey (USGS) National Wildlife Health Center (Madison, WI).

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